

15. Appendix — High-value courses (Coursera / edX / other), with effort & sequencing

Do these ADD hours to the plan? Mostly no.

The plan's ~3,000 hours already budget every phase. Each course below is one of two things:

- **⚡ Substitute** — spend a phase's *existing* budgeted hours on this course instead of the free resource → **≈ hour-neutral** (you have Lilly Coursera, so "paid" is free to you).
- **⊕ Supplement** — genuinely *extra*, optional hours — take only for breadth or if your background in that area is thin.

Effort figures are **estimates**: *full* = the whole course/specialization; *audit-select* = the audit-free subset worth watching. Use audit-select unless you want the graded work/certificate.

Default sequencing (which first)

Free plan resource FIRST ► course SECOND. The plan is intuition-first and its free resources are curated — lead with them, use a course to *consolidate/broaden*. Go **course-first** only when (a) it's the better structured option (noted per row), (b) the topic is new and you want orientation, or (c) you want a guided spine/accountability. **Never do both in full on a ▲ topic** (the plan's free resources already lead there).

Foundations & math (Phase 0-1)

- **Mathematics for Machine Learning — Imperial** · effort ~70 h full / **~40 h audit** (LA+Calc) · **⚡ first**: free (3B1B intuition) ► course as the rigorous spine. [Coursera](#)
- **Machine Learning Specialization — Andrew Ng** · ~75 h / **~30 h** · **⚡ first**: free (StatQuest is faster); take this only if you want a fuller guided pass. [Coursera](#)
- **MITx Statistics & ML (6.86x etc.) — edX** · ~120 h / ~60 h · **⊕** · optional; only if you want graded rigor beyond the plan. [edX](#)

Deep learning core (Phase 1-3)

- **Deep Learning Specialization — Andrew Ng** · ~125 h / **~70 h** · **⚡ first**: free (Karpathy for depth) ► course for breadth/tuning. [Coursera](#)
- **Practical Deep Learning for Coders — fast.ai (free)** · ~70 h · **⊕** · optional top-down complement if bottom-up theory drags. [fast.ai](#)
- **NLP Specialization — DeepLearning.AI** · ~80 h / ~30 h · **⊕** · optional; ▲ the plan's transformer resources already lead. [Coursera](#)
- **GANs Specialization — DeepLearning.AI** · ~45 h · **⊕** · optional; ▲ the plan's diffusion focus already leads. [Coursera](#)

Biomedical imaging (Phase 2-2V)

- **AI for Medicine Specialization — DeepLearning.AI** · ~55 h / **~20 h** (Diagnosis course) · **⚡ first**: free; AI-for-Medical-Diagnosis is already in the plan. [Coursera](#)

Production / MLOps (Phase 4)

- **ML Engineering for Production (MLOps) — Andrew Ng** · ~60 h / **~35 h** · **⚡ first**: COURSE (best for the MLOps vocabulary/framework) ► then the plan's free hands-on. [Coursera](#)
- **Generative AI with Large Language Models — DeepLearning.AI + AWS** · ~16 h · **⚡** · either; a clean overview to slot into Phase-4 week 1. [Coursera](#)

Agents & GenAI (Phase 6)

- **DeepLearning.AI Short Courses (free)** · ~2 h each, **~15 h** for several · **⚡ first**: COURSE (orientation) ► then build; already in the plan. [DeepLearning.AI](#)

Computer vision (Phase 7)

- **First Principles of Computer Vision — Columbia** · ~50 h full / **~18 h audit** · **⚡ first**: COURSE — the best structured classical/geometric CV anywhere; already in the plan. [Coursera](#)
- **TensorFlow Developer — DeepLearning.AI** · ~60 h · **⊕** · optional; only for TF deployment practice. [Coursera](#)

Molecular ML / drug discovery (Phase 5)

- **Bioinformatics Specialization — UC San Diego** · ~140 h full / **~40 h audit** · **⊕** · optional supplement **if your biology is thin**; audit-select. [Coursera](#)
- **Drug-discovery courses — Coursera (browse)** · ~20 h · **⊕** · optional domain context for the DMTA cycle. [Coursera](#)
- **▲** For molecular-ML *methods* (GNNs, equivariance, GFlowNets, docking/FEP, conformal UQ) the plan's free DeepChem/PyG/e3nn/RDKit/OpenMM + papers are the real curriculum — no paid course matches this yet.

Quantum / QML (Phase Q)

- **Quantum-computing intro — edX/Coursera** · ~40 h · **⊕** · optional; ▲ the plan's free IBM Quantum Learning + PennyLane already lead — skip unless you want a gentler on-ramp. [edX](#)

Net extra hours (the bottom line)

- **Substitutes (⇌): ≈ 0 net extra** — they replace free hours already budgeted (Math-for-ML, DL Spec, AI-for-Medicine, MLOps, GenAI-LLMs, First-Principles-of-CV, the short courses). Swapping in a course changes *what* you watch, not *how many* hours.
- **Supplements (⊕): all optional.** The two most worth it for you: **Bioinformatics (~40 h audit, only if biology is thin)** and **fast.ai (~70 h, only if you want a code-first complement)** — everything else is ▲-covered by the free plan. Adding just those two ≈ ~110 h extra, spread across the ~2-year window (or absorbed by the weekend buffer).
- **Rule of thumb:** treat the plan's free resources as the spine, this appendix as hour-neutral **swap-ins (⇌)** plus a few optional **supplements (⊕)**. Don't inflate the plan — the ▲ topics are already best-served free.

How to use

1. **Audit first** (free); go graded (or use Lilly Coursera) only for the assignments or a certificate that helps a specific application.
2. **Don't double-spend on ▲ topics.**
3. **Best paid-value picks for you:** Mathematics-for-ML (Imperial), MLOps (Ng), First-Principles-of-CV (Columbia) — and, if biology is thin, UCSD Bioinformatics.